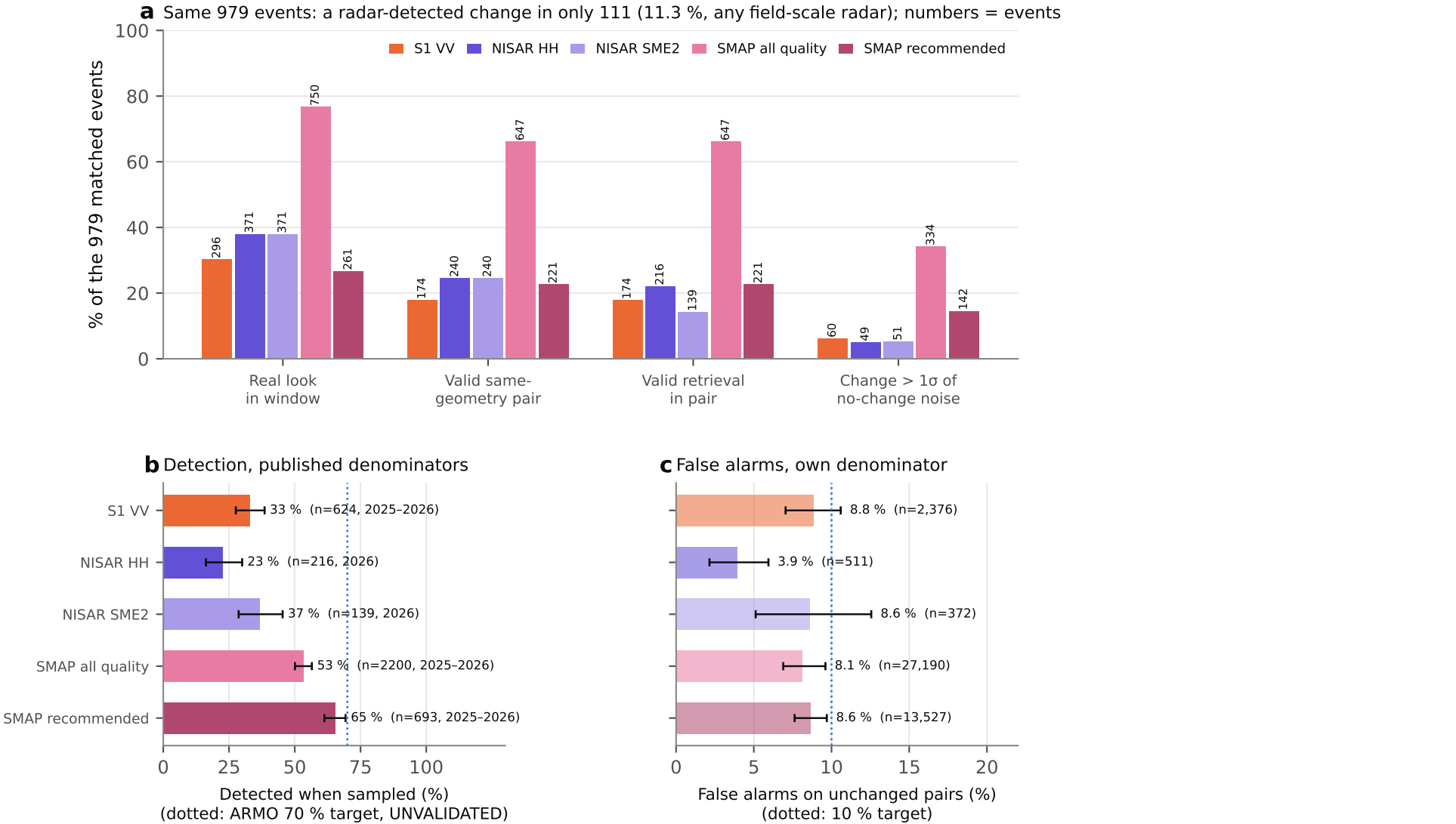

OBSERVED. (a) 979 USCRN wetting events, 107 stations, storms 17 Jun–31 Oct 2026; S1/NISAR looks = real acquisitions; SMAP look = retrieval in the window (any quality vs recommended, retrieval_qual_flag 0). Pair = same orbit/track/overpass, ground-matched, no rain in the 3 h before. Non-nested events (valid pair, no catalogue look): S1 VV 1, NISAR HH 4, NISAR SME2 4, SMAP all quality 8, SMAP recommended 4. (b, c) Different cohorts and periods per sensor; 95 % CIs = station-cluster bootstrap. False alarm = probability of a > 1σ rise on an unchanged pair (not a false-alarm ratio).